

# Diffie-Hellman Key Exchange

# Diffie-Hellman Key Exchange

- first public-key type scheme proposed
- by Diffie & Hellman in 1976 along with the exposition of public key concepts
  - note: now know that Williamson (UK CESG) secretly proposed the concept in 1970
- is a practical method for public exchange of a secret key
- used in a number of commercial products

# Diffie-Hellman Key Exchange

- a public-key distribution scheme
  - cannot be used to exchange an arbitrary message
  - rather it can establish a common key
  - known only to the two participants
- value of key depends on the participants (and their private and public key information)
- based on exponentiation in a finite (Galois) field (modulo a prime or a polynomial) - easy
- security relies on the difficulty of computing discrete logarithms (similar to factoring) – hard

$$y = a^x$$

# Primitive Root

- primitive root  $a$  is a number whose powers successively generate all the elements mod  $q$  except 0, e.g.  $q=18$

| $a$ | $a^2$ | $a^3$ | $a^4$ | $a^5$ | $a^6$ | $a^7$ | $a^8$ | $a^9$ | $a^{10}$ | $a^{11}$ | $a^{12}$ | $a^{13}$ | $a^{14}$ | $a^{15}$ | $a^{16}$ | $a^{17}$ | $a^{18}$ |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1   | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1     | 1        | 1        | 1        | 1        | 1        | 1        | 1        | 1        | 1        |
| 2   | 4     | 8     | 16    | 13    | 7     | 14    | 9     | 18    | 17       | 15       | 11       | 3        | 6        | 12       | 5        | 10       | 1        |
| 3   | 9     | 8     | 5     | 15    | 7     | 2     | 6     | 18    | 16       | 10       | 11       | 14       | 4        | 12       | 17       | 13       | 1        |
| 4   | 16    | 7     | 9     | 17    | 11    | 6     | 5     | 1     | 4        | 16       | 7        | 9        | 17       | 11       | 6        | 5        | 1        |
| 5   | 6     | 11    | 17    | 9     | 7     | 16    | 4     | 1     | 5        | 6        | 11       | 17       | 9        | 7        | 16       | 4        | 1        |
| 6   | 17    | 7     | 4     | 5     | 11    | 9     | 16    | 1     | 6        | 17       | 7        | 4        | 5        | 11       | 9        | 16       | 1        |
| 7   | 11    | 1     | 7     | 11    | 1     | 7     | 11    | 1     | 7        | 11       | 1        | 7        | 11       | 1        | 7        | 11       | 1        |
| 8   | 7     | 18    | 11    | 12    | 1     | 8     | 7     | 18    | 11       | 12       | 1        | 8        | 7        | 18       | 11       | 12       | 1        |
| 9   | 5     | 7     | 6     | 16    | 11    | 4     | 17    | 1     | 9        | 5        | 7        | 6        | 16       | 11       | 4        | 17       | 1        |
| 10  | 5     | 12    | 6     | 3     | 11    | 15    | 17    | 18    | 9        | 14       | 7        | 13       | 16       | 8        | 4        | 2        | 1        |
| 11  | 7     | 1     | 11    | 7     | 1     | 11    | 7     | 1     | 11       | 7        | 1        | 11       | 7        | 1        | 11       | 7        | 1        |
| 12  | 11    | 18    | 7     | 8     | 1     | 12    | 11    | 18    | 7        | 8        | 1        | 12       | 11       | 18       | 7        | 8        | 1        |
| 13  | 17    | 12    | 4     | 14    | 11    | 10    | 16    | 18    | 6        | 2        | 7        | 15       | 5        | 8        | 9        | 3        | 1        |
| 14  | 6     | 8     | 17    | 10    | 7     | 3     | 4     | 18    | 5        | 13       | 11       | 2        | 9        | 12       | 16       | 15       | 1        |
| 15  | 16    | 12    | 9     | 2     | 11    | 13    | 5     | 18    | 4        | 3        | 7        | 10       | 17       | 8        | 6        | 14       | 1        |
| 16  | 9     | 11    | 5     | 4     | 7     | 17    | 6     | 1     | 16       | 9        | 11       | 5        | 4        | 7        | 17       | 6        | 1        |
| 17  | 4     | 11    | 16    | 6     | 7     | 5     | 9     | 1     | 17       | 4        | 11       | 16       | 6        | 7        | 5        | 9        | 1        |
| 18  | 1     | 18    | 1     | 18    | 1     | 18    | 1     | 18    | 1        | 18       | 1        | 18       | 1        | 18       | 1        | 18       | 1        |

# Diffie-Hellman Setup

- all users agree on global parameters:
  - ① – large prime integer or polynomial  $q$
  - ② –  $a$  being a primitive root mod  $q$
- each user (eg. A) generates their key
  - chooses a secret key (number):  $x_A < q$
  - compute their **public key**:  $y_A = a^{x_A} \bmod q$
- each user makes public that key  $y_A$

*a is always  
given in question  
always given  
in questions*

*secret key must be less than  $q$*

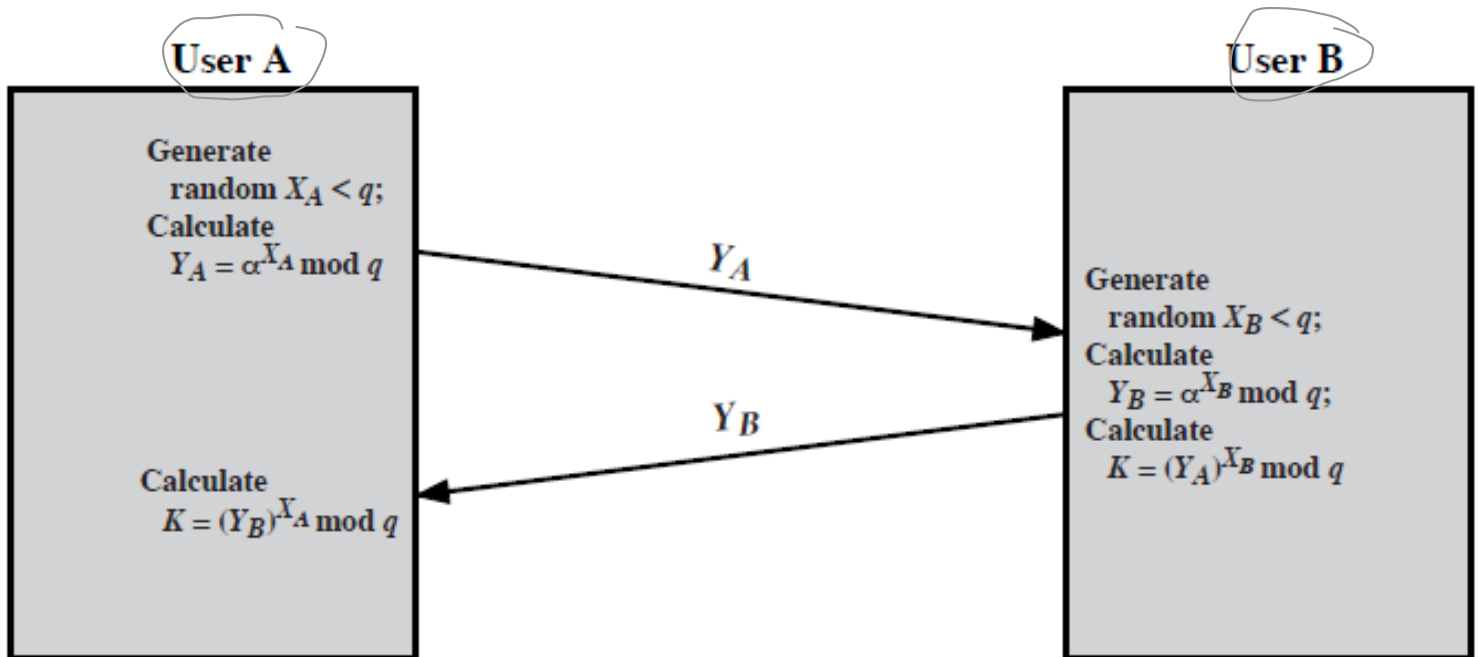
# Diffie-Hellman Key Exchange

- **shared session key** for users A & B is  $K_{AB}$ : *shared key*  
$$K_{AB} = a^{x_A \cdot x_B} \bmod q$$
$$= y_A^{x_B} \bmod q \quad (\text{which } \mathbf{B} \text{ can compute})$$
$$= y_B^{x_A} \bmod q \quad (\text{which } \mathbf{A} \text{ can compute})$$
- $K_{AB}$  is used as session key in private-key encryption scheme between Alice and Bob
- if Alice and Bob subsequently communicate, they will have the **same** key as before, unless they choose new public-keys
- attacker needs an  $x$ , must solve discrete log

$$y_A = a^{x_A} \bmod q$$

$$y_B = a^{x_B} \bmod q$$

# Diffie-Hellman Key Exchange



# Diffie-Hellman Example

- users Alice & Bob who wish to swap keys:
- agree on prime  $q=353$  and  $a=3$
- select random secret keys:

– A chooses  $x_A=97$ , B chooses  $x_B=233$

- compute respective public keys:

–  $y_A = \underset{\text{Alice}}{\overset{\text{the random key } a}{3}}^{\underline{97}} \bmod 353 = 40 \quad (\text{Alice})$

–  $y_B = \underset{\text{Bob}}{\overset{a}{3}}^{\underline{233}} \bmod 353 = 248 \quad (\text{Bob})$

- compute shared session key as: \_\_\_\_\_

–  $K_{AB} = y_B^{x_A} \bmod 353 = 248^{97} = ? \quad (\text{Alice})$

–  $K_{AB} = y_A^{x_B} \bmod 353 = 40^{233} = ? \quad (\text{Bob})$



*results  
should be  
the same*

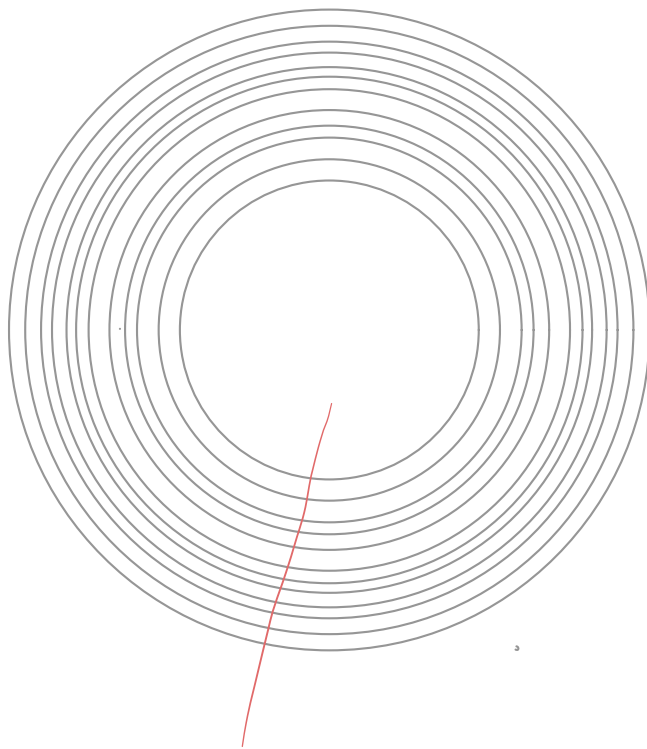
Given

$$q=23, a=9$$

Alice choose  $x=4$

Bob choose  $y=3$

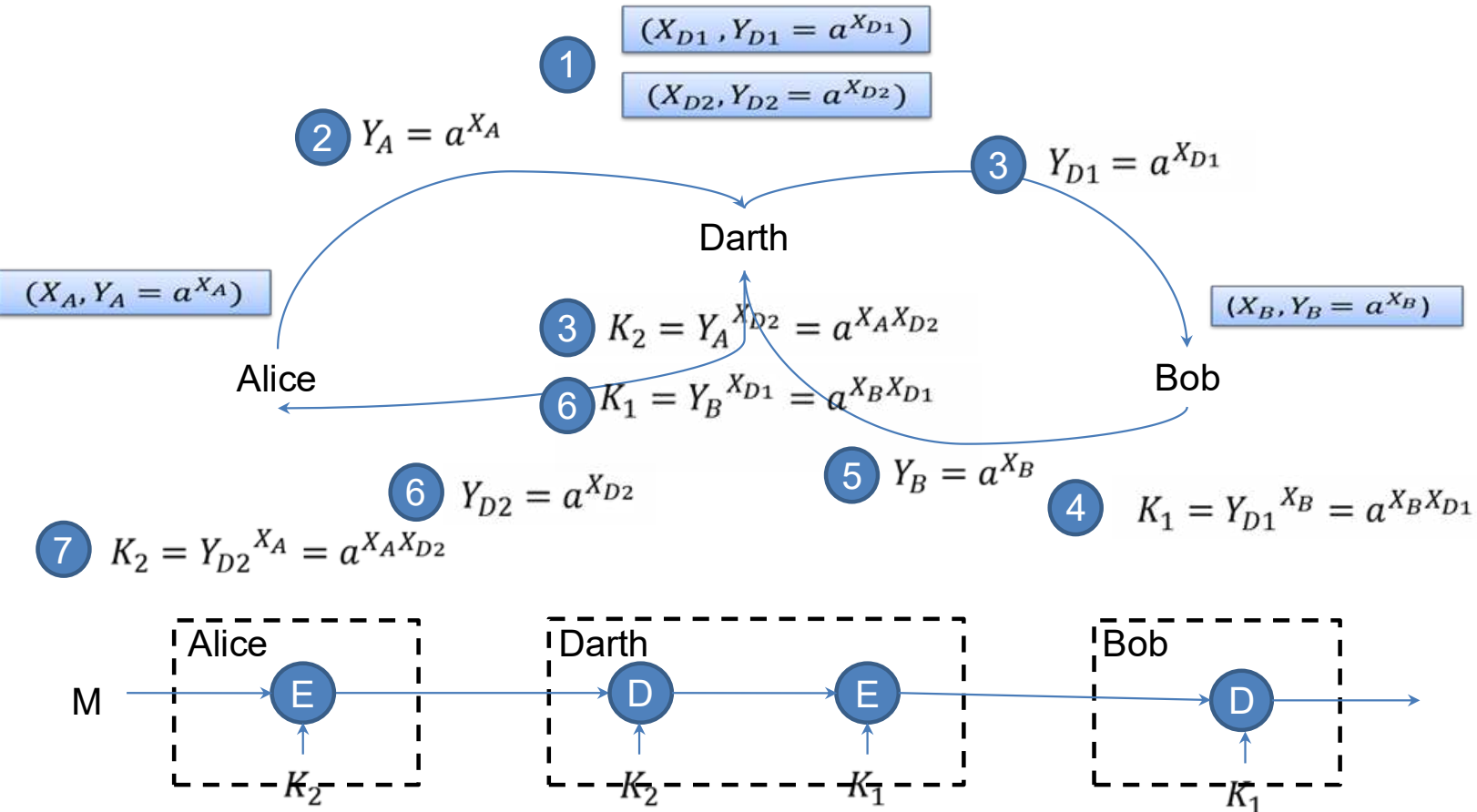
✓



# Key Exchange Protocols

- users could create random private/public D-H keys each time they communicate
- users could create a known private/public D-H key and publish in a directory, then consulted and used to securely communicate with them
- both of these are vulnerable to a meet-in-the-Middle Attack
- authentication of the keys is needed

# Man-in-the-Middle Attack



# Man-in-the-Middle Attack



1. Darth prepares by creating two private / public keys
  2. Alice transmits her public key to Bob
  3. Darth intercepts this and transmits his first public key to Bob. Darth also calculates a shared key with Alice
  4. Bob receives the public key and calculates the shared key (with Darth instead of Alice)
  5. Bob transmits his public key to Alice
  6. Darth intercepts this and transmits his second public key to Alice. Darth calculates a shared key with Bob
  7. Alice receives the key and calculates the shared key (with Darth instead of Bob)
- Darth can then intercept, decrypt, re-encrypt, forward all messages between Alice & Bob